VII SEMESTER (OPEN ELECTIVE III)	L	T	P	C
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20EC7O01 - INTRODUCTION TO GLOBAL POSITIONING SYSTEMS				

COURSE OUTCOMES:
After completion of this course, the students are able to,
CO1: Describe global navigation satellite systems (K1)
CO2: Understand GNSS Satellite signal characteristics (K2)
CO3: Develop knowledge about the various components and equipment used in GPS systems.(K3)
CO4: Develop GNSS Receiver (K3)
CO5: Analyze the impact of various error sources on the precision of positioning. (K4)

UNIT I : FUNDAMENTALS OF SATELLITE NAVIGATION:
Concept of Ranging using Time of arrival Measurements: Two-Dimensional Position Determination, Principle of Position Determination via Satellite-Generated Ranging signals, Fundamentals of satellite orbits: Orbital Mechanics, Constellation Design, Positioning determination using Ranging codes: Determining Satellite-to-User Range, Indian Developed GNSS- Indian Regional Navigation Satellite System (IRNSS) : NavIC and its applications, *GPS*-Aided Geo-Augmented *Navigation (GAGAN)*

UNIT II : GLOBAL POSITIONING SYSTEM SEGMENTS:
Space Segment Description: GPS Satellite Constellation Description, Constellation Design Guidelines, Space Segment Phased Development, Control Segment: Current Configuration, CS Planned Upgrades , User Segment: GPS Set Characteristics, GPS Receiver Selection

UNIT-III : GPS SATELLITE SIGNAL CHARACTERISTICS:
Modulations for Satellite Navigation: Modulation Types, Multiplexing Techniques, Signal Models and Characteristics, Legacy GPS Signals: Frequencies and Modulation Format, Power Levels, Autocorrelation Functions and Power Spectral Densities, Cross-Correlation Functions and CDMA Performance, Navigation Message Format.

UNIT-IV : GNSS RECEIVER:
Acquisition: Single Trial Detector, Tong Search Detector, M of N Search Detector, Combined Tong and M of N Search Detectors, FFT-Based Techniques, Direct Acquisition of GPS Military Signals, Vernier Doppler and Peak Code Search, carrier tracking, code tracking: Carrier Loop Discriminator, sequence of initial receiver operation.

UNIT-V: GNSS ERRORS: Introduction, Measurement errors: satellite clock error, ephemeris error, relative effects, atmospheric effects, receiver noise and resolution, multipath and shadowing effects, hardware bias errors, Pseudo range error budgets.

TExt Books:
1. Elliott D. Kaplan, Christopher J. Hegarty, Understanding **GPS/GNSS** principles and applications, third edition, artech house publishers, Boston, 2017
2. G S Rao, Global Navigational satellite system, Tata McGraw-Hill education private Ltd, New Delhi, 2015.

References:
1. ISRO-IRNSS-ICD-SPS-1.1, Bangalore, 2017
2. Bhatta, B. ". Global Navigation Satellite Systems: Insights Into GPS, Glonass, Galileo, Compass, and Others", BS Publications, New Delhi, 2015.

E-References:
1. https://archive.nptel.ac.in/courses/105/107/105107194/
2. https://d1.amobbs.com/bbs_upload782111/files_33/ourdev_584835O21W59.pdf